

Aptitude Assessment 1 (AT1)

STUDENT COPY

ASSESSMENT QUESTION PAPER

Test Code: **AT1** • Total Questions: **150** • Max Marks: **150**

Duration: **90 Minutes**

Format: **Multiple Choice Questions**

Student Name: _____

Roll Number: _____

Date: _____

Score: _____ /
150

Question 1

[1 Mark]

Three numbers are in the ratio of 2 : 3 : 5 and their L.C.M. is 1800. Their H.C.F. is:

A 30

B 60

C 90

D 120

Question 2

[1 Mark]

The ratio of two numbers is 4 : 5 and their H.C.F. is 6. Their L.C.M. is:

A 60

B 90

C 120

D 150

Question 3

[1 Mark]

The L.C.M. of two numbers is 72. The numbers are in the ratio 3 : 4. Then the sum of the numbers is:

A 36

B 42

C 48

D 54

Question 4

[1 Mark]

If one-fourth of one-fifth of a number is 16, then three-eighth of that number is:

A 90

B 105

C 120

D 140

Question 5

[1 Mark]

Three times the first of three consecutive odd integers is 7 more than twice the third. The third integer is:

A 15

B 17

C 19

D 21

Question 6

[1 Mark]

The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 45. What is the difference between the two digits of that number?

A 4**B** 5**C** 6**D** Cannot be determined**Question 7**

[1 Mark]

A two-digit number is such that the product of its digits is 12. When 36 is added to the number, the digits are reversed. The number is:

A 26**B** 34**C** 43**D** 62**Question 8**

[1 Mark]

The sum of the digits of a two-digit number is 13 and the difference between the digits is 5. What is the two-digit number?

A 49**B** 94**C** 58**D** Cannot be determined**Question 9**

[1 Mark]

Which one of the following numbers is exactly divisible by 11?

A 28182**B** 28185**C** 315624**D** 415623**Question 10**

[1 Mark]

If the number $532 * 714$ is completely divisible by 9, then the smallest whole number in place of * will be:

A 3**B** 4**C** 5**D** 6

Question 11

[1 Mark]

The value of a printing machine depreciates at the rate of 10% every year. If its present value is Rs. 14,580, then what was the price of the machine three years ago?

A Rs. 18,000**B** Rs. 20,000**C** Rs. 22,000**D** Rs. 25,000**Question 12**

[1 Mark]

A student erroneously multiplied a number by $\frac{3}{4}$ instead of $\frac{4}{3}$. What is the percentage error in the calculation?

A 36.5%**B** 43.75%**C** 48.2%**D** 56.25%**Question 13**

[1 Mark]

Find the value of 20% of 15% of Rs. 1,200.

A Rs. 28**B** Rs. 32**C** Rs. 36**D** Rs. 40**Question 14**

[1 Mark]

The population of a town is decreasing at a rate of 15% per annum. If the population two years ago was 40,000, what is the present population?

A 27,500**B** 28,900**C** 30,200**D** 32,400**Question 15**

[1 Mark]

Suresh spends 65% of his salary and deposits 20% of his salary in the bank. If he is left with Rs. 3,000, what is his monthly salary?

A Rs. 18,000**B** Rs. 20,000**C** Rs. 22,500**D** Rs. 25,000

Question 16

[1 Mark]

A fruit seller had some oranges. He sells 45% oranges and still has 495 oranges. Originally, he had:

A 800 oranges**B** 850 oranges**C** 900 oranges**D** 950 oranges**Question 17**

[1 Mark]

Three candidates contested an election and received 1,200, 4,800, and 6,000 votes respectively. What percentage of the total votes did the winning candidate get?

A 45%**B** 48%**C** 50%**D** 55%**Question 18**

[1 Mark]

The value of a CNC lathe machine depreciates at the rate of 10% per annum. If the cost of the machine at present is Rs. 200,000, then what will be its worth after 2 years?

A Rs. 158,000**B** Rs. 162,000**C** Rs. 166,000**D** Rs. 172,000**Question 19**

[1 Mark]

Passing marks in an examination are 40%. If a candidate gets 85 marks and fails by 15 marks, the maximum marks in the examination are:

A 200**B** 225**C** 250**D** 300**Question 20**

[1 Mark]

The price of an article was first decreased by 25% and then increased by 20%. What was the overall percent decrease or increase?

A 5% increase**B** 10% decrease**C** 8% decrease**D** No change

Question 21

[1 Mark]

A train travelled a certain distance at a uniform speed. Had the speed been 6 km/h more, it would have needed 4 hours less. Had the speed been 6 km/h less, it would have needed 6 hours more. The distance travelled by the train is:

A 720 km**B** 540 km**C** 640 km**D** 840 km**Question 22**

[1 Mark]

A boat's speed in still water is 20 km/h, while the river flows with a speed of 5 km/h. If the time taken to cover a certain distance upstream is 4 hours more than downstream, find the distance.

A 120 km**B** 150 km**C** 180 km**D** 200 km**Question 23**

[1 Mark]

The speed of a boat in still water is thrice the speed of the stream. If the boat takes 18.5 sec to go to a certain place downstream, find the additional time required to cover the same distance travelling upstream:

A 18.5 sec**B** 22.5 sec**C** 37.0 sec**D** 15.0 sec**Question 24**

[1 Mark]

A boat goes 24 km upstream and 36 km downstream in 6 hours. In 8 hours, it goes 32 km upstream and 48 km downstream. Determine the speed of the boat in still water.

A 10 km/h**B** 8 km/h**C** 12 km/h**D** 14 km/h**Question 25**

[1 Mark]

In a metropolitan area, last year's tourism revenue accounted for 25% of overall revenue, with the remaining \$750,000 coming from tech manufacturing. This year, tourism revenue fell by 60%, while manufacturing rose by 150%. What percentage of this year's manufacturing revenue is the revenue from tourism?

A 4%**B** 5.33%**C** 6.67%**D** 8%

Question 26

[1 Mark]

In a school, 20% of students are below 8 years of age. The number of students above 8 years is $\frac{2}{3}$ of the students of 8 years of age, which is 60. What is the total number of students in the school?

A 125**B** 150**C** 175**D** 200**Question 27**

[1 Mark]

In an election between two candidates, one got 60% of the total valid votes, and 20% of the total votes were invalid. If the total number of votes was 8,000, the number of valid votes that the other candidate received was:

A 2,560**B** 3,200**C** 3,840**D** 4,200**Question 28**

[1 Mark]

The incomes of Alex, Bob, and Charlie are in the ratio 7 : 9 : 12, while their expenses are in the ratio 8 : 9 : 15. If their respective savings are Rs. 200, Rs. 450, and Rs. 150, what is Alex's income?

A Rs. 1,400**B** Rs. 1,200**C** Rs. 1,600**D** Rs. 1,800**Question 29**

[1 Mark]

A can complete a task in 12 days and B in 18 days. If they work together for 4 days, what fraction of the work remains unfinished?

A $\frac{4}{9}$ **B** $\frac{5}{9}$ **C** $\frac{1}{3}$ **D** $\frac{2}{9}$ **Question 30**

[1 Mark]

A, B, and C can complete a job in 20, 30, and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

A 15 days**B** 12 days**C** 18 days**D** 20 days

Question 31

[1 Mark]

Evaluate: $25\% \text{ of } 40 + 25\% \text{ of } 60 - 15\% \text{ of } 200 = ?$

A -5**B** 0**C** -10**D** 5**Question 32**

[1 Mark]

The population of a city increased from 2,00,000 to 3,00,000 in a decade. The average percentage increase of population per year is:

A 5%**B** 6%**C** 7.5%**D** 8%**Question 33**

[1 Mark]

If A:B is 1:2, B:C is 3:4, and C:D is 2:3, find A:B:C:D.

A 3:6:8:12**B** 2:4:6:9**C** 1:2:3:4**D** 3:5:7:9**Question 34**

[1 Mark]

The ratio of two numbers is 7 : 9. If the sum of the numbers is 256, find the difference between the numbers.

A 32**B** 28**C** 36**D** 40**Question 35**

[1 Mark]

The ages of three siblings are in the ratio 4 : 7 : 9. If the sum of their ages is 40 years, find the difference between the eldest and the youngest in months.

A 120 months**B** 108 months**C** 96 months**D** 132 months**Question 36**

[1 Mark]

Two numbers are respectively 30% and 60% more than a third number. The ratio of the two numbers is:

A 13 : 16**B** 3 : 4**C** 11 : 14**D** 15 : 19

Question 37

[1 Mark]

A sum of money is divided among A, B, C, and D in the ratio 4 : 3 : 6 : 2. If C receives Rs. 1,200 more than D, what is B's share?

A Rs. 900**B** Rs. 750**C** Rs. 1,050**D** Rs. 1,200**Question 38**

[1 Mark]

If $0.6 : x :: 3 : 5$, then x is equal to:

A 1.0**B** 1.2**C** 0.8**D** 1.5**Question 39**

[1 Mark]

The sum of three numbers is 105. If the ratio of the first to the second is 2 : 3 and that of the second to the third is 4 : 5, then the second number is:

A 36**B** 24**C** 45**D** 48**Question 40**

[1 Mark]

If Rs. 940 is divided into three parts proportional to $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$, then the first part is:

A Rs. 433.85**B** Rs. 420**C** Rs. 450**D** Rs. 400**Question 41**

[1 Mark]

The average weight of A, B, and C is 50 kg. If the average weight of A and B is 45 kg and that of B and C is 48 kg, then the weight of B is:

A 36 kg**B** 38 kg**C** 40 kg**D** 42 kg

Question 42

[1 Mark]

The average weight of 40 students is 52 kg. It was found later that one weight of 35 kg was misread as 75 kg. What is the correct average?

A 51 kg**B** 50.5 kg**C** 51.5 kg**D** 50 kg**Question 43**

[1 Mark]

The average age of 25 students is 12 years. If the teacher's age is included, the average increases by 1 year. What is the age of the teacher?

A 38 years**B** 36 years**C** 40 years**D** 42 years**Question 44**

[1 Mark]

The average marks of 150 candidates was 40. If the average of passed candidates was 45 and failed candidates was 20, the number of candidates who passed is:

A 120**B** 110**C** 125**D** 130**Question 45**

[1 Mark]

Five years ago, the ages of A and B were in the ratio 2 : 3. At present, the ratio is 3 : 4. Find the present age of A:

A 15 years**B** 18 years**C** 20 years**D** 24 years**Question 46**

[1 Mark]

Two years ago, the ratio of Ron and Sam's ages was 3 : 4. Six years hence, this ratio will become 5 : 6. How old is Sam now?

A 18 years**B** 16 years**C** 20 years**D** 22 years

Question 47

[1 Mark]

Present ages of Rahul and Amit are in the ratio 6 : 5. Four years hence, their ages will be in the ratio 7 : 6. What is Amit's present age?

A 20 years**B** 24 years**C** 28 years**D** 32 years**Question 48**

[1 Mark]

Four years ago, a father was 6 times as old as his son. At present, the father is 4 times as old as his son. What are their present ages?

A 40, 10**B** 36, 9**C** 48, 12**D** 32, 8**Question 49**

[1 Mark]

Eight years ago, P was one-third of Q in age. If the ratio of their present ages is 3 : 5, what is the sum of their present ages?

A 32 years**B** 40 years**C** 48 years**D** 56 years**Question 50**

[1 Mark]

Ten years ago, P was half of Q in age. If the ratio of their present ages is 3 : 4, what is the total of their present ages?

A 35 years**B** 40 years**C** 45 years**D** 50 years**Question 51**

[1 Mark]

A bank offers 6% compound interest calculated on a half-yearly basis. A customer deposits Rs. 2,000 each on 1st January and 1st July of a year. At the end of the year, the amount gained by way of interest is:

A Rs. 181.80**B** Rs. 175.00**C** Rs. 192.40**D** Rs. 160.00

Question 52

[1 Mark]

The difference between simple and compound interests compounded annually on a sum for 2 years at 5% per annum is Rs. 15. The sum is:

A Rs. 6,000**B** Rs. 5,000**C** Rs. 7,200**D** Rs. 8,000**Question 53**

[1 Mark]

A father is aged three times more than his son Ronit. After 8 years, he would be two and a half times of Ronit's age. After further 8 years, how many times would he be of Ronit's age?

A 2 times**B** 2.25 times**C** 2.5 times**D** 3 times**Question 54**

[1 Mark]

A father said to his son, 'I was as old as you are at the present at the time of your birth'. If the father's age is 42 years now, the son's age five years back was:

A 16 years**B** 18 years**C** 21 years**D** 26 years**Question 55**

[1 Mark]

A card is drawn from a well-shuffled pack of 52 playing cards. What is the probability of getting a queen of clubs or a king of hearts?

A $1/26$ **B** $1/52$ **C** $2/13$ **D** $1/13$ **Question 56**

[1 Mark]

From a group of 6 men and 5 women, a committee of 4 persons is to be formed so that at least 2 men are on the committee. In how many ways can this be done?

A 265 ways**B** 280 ways**C** 295 ways**D** 310 ways

Question 57

[1 Mark]

In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels always come together?

A 144**B** 120**C** 240**D** 720**Question 58**

[1 Mark]

Seeta, Reeta and Geeta spend 12%, 14% and 16% of their monthly salary on travelling respectively and each saves half of the remaining amount. Seeta and Geeta have identical monthly salaries. Seeta's monthly saving is Rs. 360 more than Geeta's, and the travelling expenditure of Seeta and Reeta together is Rs. 1240 more than that of Geeta. The monthly salary of Reeta is how much less than that of Seeta?

A Rs. 4,000 less**B** Rs. 3,000 less**C** Rs. 2,000 less**D** Rs. 5,000 less**Question 59**

[1 Mark]

Based on the salary data of Seeta, Reeta and Geeta (Salaries: Seeta = Rs. 18,000, Reeta = Rs. 14,000, Geeta = Rs. 18,000; travel spends 12%, 14%, 16%; savings = half of remainder), what is the sum of the savings of all three friends together?

A Rs. 21,500**B** Rs. 22,000**C** Rs. 23,500**D** Rs. 24,000**Question 60**

[1 Mark]

Based on the salary data of Seeta, Reeta and Geeta (Total monthly savings = Rs. 21,500, Total monthly salary = Rs. 50,000), the total monthly savings of the three friends together is what percentage of their total monthly salary?

A 43%**B** 45%**C** 48%**D** 50%**Question 61**

[1 Mark]

Rajeev's age after 15 years will be 5 times his age 5 years back. What is the present age of Rajeev?

A 10 years**B** 8 years**C** 12 years**D** 15 years

Question 62

[1 Mark]

The ratio between Rahul and Deepak's ages is 4 : 3. After 6 years, Rahul's age will be 26 years. What is the present age of Deepak?

A 15 years**B** 12 years**C** 18 years**D** 20 years**Question 63**

[1 Mark]

The ratio of a father's age to his son's age is 4 : 1. The product of their ages is 196. What will be the ratio of their ages after 5 years?

A 11 : 4**B** 5 : 2**C** 11 : 6**D** 13 : 5**Question 64**

[1 Mark]

A is two years older than B, who is twice as old as C. If the total of the ages of A, B, and C is 27, how old is B?

A 10 years**B** 8 years**C** 12 years**D** 14 years**Question 65**

[1 Mark]

The average age of Ram and Shyam is 65 years. The average age of Ram, Shyam, and John is 53 years. What is the age of John?

A 29 years**B** 27 years**C** 31 years**D** 33 years**Question 66**

[1 Mark]

A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share of the remaining profit is Rs. 855, what is the total profit?

A Rs. 1,500**B** Rs. 1,425**C** Rs. 1,600**D** Rs. 1,750

Question 67

[1 Mark]

A starts a business with Rs. 3,500 and after 5 months, B joins as a partner. After a full year, the profit is divided in the ratio 2 : 3. What was B's capital contribution?

A Rs. 9,000**B** Rs. 8,500**C** Rs. 7,500**D** Rs. 8,000**Question 68**

[1 Mark]

Simran started a business investing Rs. 50,000. After six months, Nanda joined with Rs. 80,000. After 3 years, they earned a total profit of Rs. 24,500. What was Simran's share in the profit?

A Rs. 10,500**B** Rs. 9,423**C** Rs. 12,500**D** Rs. 14,000**Question 69**

[1 Mark]

If Rs. 100,000 and Rs. 150,000 are invested by A and B respectively, and the annual profit is Rs. 24,000, what is A's share?

A Rs. 9,600**B** Rs. 8,000**C** Rs. 10,500**D** Rs. 12,000**Question 70**

[1 Mark]

X and Y invested in a business and divided profits in the ratio 2 : 3. If X invested Rs. 40,000, find the capital invested by Y:

A Rs. 60,000**B** Rs. 50,000**C** Rs. 75,000**D** Rs. 80,000**Question 71**

[1 Mark]

A, B, and C started a business by investing Rs. 25,000, Rs. 30,000, and Rs. 35,000 respectively. Find the share of B out of an annual profit of Rs. 187,200:

A Rs. 62,400**B** Rs. 52,000**C** Rs. 72,800**D** Rs. 65,400

Question 72

[1 Mark]

Rs. 1,400 is divided among Amla, Bimla, and Simla so that Amla receives half as much as Bimla, and Bimla receives half as much as Simla. What is Simla's share?

A Rs. 800**B** Rs. 400**C** Rs. 600**D** Rs. 700**Question 73**

[1 Mark]

Smith and Kate started a venture investing Rs. 84,000 and Rs. 28,000 respectively. In what ratio should the profit earned after 2 years be divided between Smith and Kate?

A 3 : 1**B** 2 : 1**C** 4 : 1**D** 5 : 2**Question 74**

[1 Mark]

Sham and Ram gained profit divided in the ratio 2 : 3. If Sham invested Rs. 40,000, what was the capital invested by Ram?

A Rs. 60,000**B** Rs. 50,000**C** Rs. 75,000**D** Rs. 80,000**Question 75**

[1 Mark]

Harry, John, and Smith start a business investing Rs. 27,000, Rs. 72,000, and Rs. 81,000 respectively. At the end of the year, Smith earns a profit share of Rs. 36,000. Find the total profit:

A Rs. 80,000**B** Rs. 75,000**C** Rs. 90,000**D** Rs. 85,000**Question 76**

[1 Mark]

In what ratio should two varieties of sugar of Rs. 18/kg and Rs. 24/kg be mixed together to get a mixture whose cost is Rs. 20/kg?

A 2 : 1**B** 1 : 2**C** 3 : 1**D** 4 : 1

Question 77

[1 Mark]

Two vessels A and B contain spirit and water in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which these mixtures must be combined to obtain a new mixture containing spirit and water in the ratio 8 : 5:

A 7 : 9**B** 5 : 7**C** 6 : 7**D** 8 : 9**Question 78**

[1 Mark]

How much water must be added to 40 litres of milk at the cost price of Rs. 3.50 per litre so that the cost of the diluted milk is reduced to Rs. 2.00 per litre?

A 30 litres**B** 25 litres**C** 35 litres**D** 20 litres**Question 79**

[1 Mark]

A dishonest milkman professes to sell milk at cost price but mixes it with water and thereby gains 25%. What is the percentage of water in the final mixture?

A 20%**B** 25%**C** 15%**D** 30%**Question 80**

[1 Mark]

In a 729-litre mixture of milk and water, the ratio of milk to water is 7 : 2. To obtain a new mixture containing milk and water in the ratio 7 : 3, what amount of water must be added?

A 81 litres**B** 75 litres**C** 90 litres**D** 65 litres**Question 81**

[1 Mark]

In what ratio must wheat A at Rs. 10.50 per kg be mixed with wheat B at Rs. 12.30 per kg so that the mixture is worth Rs. 11.00 per kg?

A 13 : 5**B** 11 : 5**C** 12 : 7**D** 14 : 5

Question 82

[1 Mark]

In what ratio must a shopkeeper mix Peas at Rs. 16/kg and Soybeans at Rs. 25/kg to obtain a mixture worth Rs. 19.50/kg?

A 11 : 7**B** 9 : 5**C** 7 : 5**D** 12 : 8**Question 83**

[1 Mark]

Two containers P and Q contain milk and water in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which these two mixtures should be combined to produce a mixture R with ratio 8 : 5:

A 7 : 9**B** 5 : 6**C** 4 : 9**D** 9 : 7**Question 84**

[1 Mark]

A container contains 30 litres of pure milk. From this container, 5 litres of milk is taken out and replaced by water. This process is repeated further two times. How much pure milk remains in the container?

A 17.36 litres**B** 18.45 litres**C** 21.87 litres**D** 23.44 litres**Question 85**

[1 Mark]

What will be the simple interest on Rs. 80,000 at 10% per annum for 9 months?

A Rs. 6,000**B** Rs. 8,000**C** Rs. 7,500**D** Rs. 5,400**Question 86**

[1 Mark]

At what rate percent per annum will a sum of money double itself in 20 years at simple interest?

A 5%**B** 4%**C** 6%**D** 8%

Question 87

[1 Mark]

A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The principal sum is:

A Rs. 698**B** Rs. 650**C** Rs. 690**D** Rs. 700**Question 88**

[1 Mark]

A sum of Rs. 12,500 amounts to Rs. 15,500 in 4 years at simple interest. What is the annual rate of interest?

A 6%**B** 4%**C** 5%**D** 7%**Question 89**

[1 Mark]

A, B, and C invest Rs. 20,000, Rs. 25,000, and Rs. 35,000 in a business. Find the share of B out of a total profit of Rs. 128,000:

A Rs. 40,000**B** Rs. 32,000**C** Rs. 45,000**D** Rs. 48,000**Question 90**

[1 Mark]

Rs. 2,100 is divided among X, Y, and Z such that X receives half as much as Y, and Y receives half as much as Z. What is Z's share?

A Rs. 1,200**B** Rs. 600**C** Rs. 900**D** Rs. 1,000**Question 91**

[1 Mark]

P and Q start a business with Rs. 45,000 and Rs. 15,000 respectively. In what ratio should the profit after 1 year be divided?

A 3 : 1**B** 2 : 1**C** 4 : 1**D** 5 : 2

Question 92

[1 Mark]

A and B divide profit in the ratio 3 : 5. If A invested Rs. 45,000, what was B's investment?

A Rs. 75,000**B** Rs. 60,000**C** Rs. 80,000**D** Rs. 90,000**Question 93**

[1 Mark]

Three partners invest in the ratio 2 : 5 : 8. The third partner earns Rs. 48,000 as profit. What is the total profit?

A Rs. 90,000**B** Rs. 80,000**C** Rs. 96,000**D** Rs. 100,000**Question 94**

[1 Mark]

A shopkeeper sells an item for Rs. 240 at a loss of 20%. What is the cost price of the article?

A Rs. 300**B** Rs. 280**C** Rs. 320**D** Rs. 350**Question 95**

[1 Mark]

The selling price of 10 candles is equal to the cost price of 12 candles. What is the gain percent?

A 20%**B** 15%**C** 25%**D** 18%**Question 96**

[1 Mark]

The selling price of 40 articles is equal to the cost price of 35 articles. Find the profit or loss percent:

A 12.5% loss**B** 12.5% gain**C** 14.2% loss**D** 10% loss**Question 97**

[1 Mark]

If a retailer marks up goods by 25% and then offers a 20% discount on the marked price, what is the net profit or loss percentage?

A 0% (No profit no loss)**B** 5% gain**C** 5% loss**D** 2% gain

Question 98

[1 Mark]

A merchant made a profit of 25% by selling an outfit for Rs. 5,000. What was the cost price of the outfit?

A Rs. 4,000**B** Rs. 3,750**C** Rs. 4,200**D** Rs. 4,500**Question 99**

[1 Mark]

In what ratio must rice at Rs. 24/kg be mixed with premium rice at Rs. 34/kg to obtain a blend worth Rs. 30/kg?

A 2 : 3**B** 3 : 2**C** 1 : 2**D** 3 : 4**Question 100**

[1 Mark]

In what ratio should cashews at Rs. 400/kg be mixed with almonds at Rs. 600/kg to achieve a blend priced at Rs. 480/kg?

A 3 : 2**B** 2 : 3**C** 1 : 1**D** 4 : 3**Question 101**

[1 Mark]

Two containers have acid and water in ratios 3 : 1 and 5 : 3. In what ratio must they be mixed to obtain a 2 : 1 ratio?

A 1 : 1**B** 2 : 1**C** 3 : 2**D** 1 : 2**Question 102**

[1 Mark]

A vat contains 40 litres of chemical solution. 4 litres are drawn off and replaced with water. This operation is repeated twice more. How much of the original chemical remains?

A 29.16 litres**B** 28.00 litres**C** 30.24 litres**D** 27.50 litres

Question 103

[1 Mark]

Find the simple interest on Rs. 45,000 at 8% per annum for 6 months:

A Rs. 1,800**B** Rs. 2,400**C** Rs. 1,600**D** Rs. 2,100**Question 104**

[1 Mark]

At what simple interest rate per annum will an initial deposit double in 10 years?

A 10%**B** 8%**C** 12%**D** 15%**Question 105**

[1 Mark]

A sum at simple interest amounts to Rs. 920 in 2 years and Rs. 1,040 in 3 years. What is the initial principal?

A Rs. 680**B** Rs. 720**C** Rs. 640**D** Rs. 700**Question 106**

[1 Mark]

A principal of Rs. 8,000 yields Rs. 1,920 interest over 3 years. What is the rate of interest per annum?

A 8%**B** 6%**C** 7.5%**D** 9%**Question 107**

[1 Mark]

Find the missing term in the sequence: ABC, ABC2, AB2C2, ____, A2B2C3

A A2B2C2**B** ABC3**C** AB3C2**D** A3B2C2**Question 108**

[1 Mark]

Find the next term in the series: BLL, DNN, FPP, ____, JTT

A HRR**B** GQQ**C** HSS**D** IRR

Question 109

[1 Mark]

Look at the series: 3, 9, 27, 81, ... What number should come next?

A 243**B** 162**C** 216**D** 324**Question 110**

[1 Mark]

Look at the sequence: 5, 4, 6, 15, 56, ... What number should come next?

A 275**B** 265**C** 280**D** 290**Question 111**

[1 Mark]

Statements: All desks are chairs. No chair is a table.

Conclusions:

(I) Some desks are tables.

(II) No table is a desk.

A Only (II) follows**B** Only (I) follows**C** Both follow**D** Neither follows**Question 112**

[1 Mark]

Given the rules:

A # B means B is 1m East of A

A \$ B means B is 1m North of A

A * B means B is 1m West of A

A @ B means B is 1m South of A

According to $P @ Q * R$, in which direction is R with respect to P?

A South-West**B** North-East**C** South-East**D** North-West**Question 113**

[1 Mark]

Using the same rules (# = East, \$ = North, * = West, @ = South):

According to $J \# K \$ L$, in which direction is L with respect to J?

A North-East**B** North-West**C** South-East**D** South-West

Question 114

[1 Mark]

Look at this series: 34, 33, 35, 34, 36, 35, ... What number should come next?

☐ A 37☐ B 36☐ C 38☐ D 34**Question 115**

[1 Mark]

Look at this series: 100, 20, 90, 25, 80, ... What number should come next?

☐ A 30☐ B 35☐ C 70☐ D 75**Question 116**

[1 Mark]

One morning after sunrise, Aarav and Varun were talking face to face. If Varun's shadow fell exactly to the left of Aarav, which direction was Aarav facing?

☐ A North☐ B South☐ C East☐ D West**Question 117**

[1 Mark]

If South-East becomes North, North-East becomes West, and so on. What will West become in this coordinate system?

☐ A South-East☐ B North-West☐ C South-West☐ D North-East**Question 118**

[1 Mark]

A clock is placed such that at 6:00 PM, the hour hand points North. In which direction will the minute hand point at 9:15 PM?

☐ A West☐ B East☐ C North☐ D South

Question 119

[1 Mark]

From home, Maya walked 20 km North, turned West and walked 15 km, turned South and walked 8 km, then turned East and walked 15 km. In which direction and distance is she from her home?

☐ A 12 km North☐ B 12 km South☐ C 15 km North☐ D 8 km East**Question 120**

[1 Mark]

Pointing to a photograph of a boy, Ramesh said, 'He is the son of the only son of my mother.' How is Ramesh related to the boy?

☐ A Father☐ B Uncle☐ C Brother☐ D Grandfather**Question 121**

[1 Mark]

Pointing to a photograph, Meera said, 'He is the son of the only son of my paternal grandfather.' How is the man in the photograph related to Meera?

☐ A Brother☐ B Father☐ C Cousin☐ D Uncle**Question 122**

[1 Mark]

Consider the triangular pattern where bottom numbers (A, B, C) relate to the apex number:

Row 1: (5, 4, 7) -> 8

Row 2: (6, 9, 5) -> 10

Row 3: (3, 7, 2) -> ?

What number replaces the question mark?

☐ A 6☐ B 8☐ C 10☐ D 12**Question 123**

[1 Mark]

Statements: All pens are papers. All papers are folders.

Conclusions:

(I) Some folders are pens.

(II) All pens are folders.

☐ A Both follow☐ B Only (I) follows☐ C Only (II) follows☐ D Neither follows

Question 124

[1 Mark]

Statements: All birds are fliers. Some kites are fliers.

Conclusions:

(I) All birds are kites.

(II) Some kites are birds.

A Neither follows

B Only (I) follows

C Only (II) follows

D Both follow

Question 125

[1 Mark]

Look at the sequence: 589654237, 89654237, 8965423, 965423, ?

What comes next?

A 96542

B 96543

C 65423

D 96785

Question 126

[1 Mark]

Look at the sequence: 11, 10, ?, 100, 1001, 1000, 10001

What term replaces the question mark?

A 101

B 110

C 111

D 1001

Question 127

[1 Mark]

Data Interpretation: A family's monthly budget is Rs. 60,000 allocated as: Food (20%), Savings (15%), Clothing (15%), Housing (10%), Education (10%), Transport (15%), Miscellaneous (15%). How much is spent on Food?

A Rs. 12,000

B Rs. 10,000

C Rs. 14,000

D Rs. 15,000

Question 128

[1 Mark]

From the budget (Rs. 60,000 total: Clothing 15%, Housing 10%, Transport 15%), how much more money is spent on Clothing and Housing together than on Transport?

A Rs. 6,000

B Rs. 4,500

C Rs. 7,500

D Rs. 9,000

Question 129

[1 Mark]

From the budget (Food = 20%, Education = 10%), what percentage of the amount spent on food is spent on education?

A 50%**B** 25%**C** 40%**D** 60%**Question 130**

[1 Mark]

In a pie chart representing sales by segment: Segment A (30%), Segment B (25%), Segment C (35%), Segment D (10%). What is the central angle corresponding to Segment A?

A 108°**B** 90°**C** 126°**D** 120°**Question 131**

[1 Mark]

Solve for A, B, C in the cryptarithmic puzzle: $ABC + CBA = 444$ where distinct letters represent distinct non-zero digits. A possible solution is:

A A=1, B=2, C=3**B** A=2, B=4, C=2**C** A=3, B=1, C=1**D** A=1, B=7, C=2**Question 132**

[1 Mark]

Solve for D, E, F in: $DEF - FED = 297$ where $D=5$, $F=2$. What is the value of E?

A Any single digit 0-9**B** E=4**C** E=7**D** E=0**Question 133**

[1 Mark]

Solve for X, Y, Z in: $XYZ + XYZ = 468$ with unique digits. Find X, Y, Z:

A X=2, Y=3, Z=4**B** X=2, Y=4, Z=2**C** X=3, Y=2, Z=4**D** X=1, Y=8, Z=4

Question 134

[1 Mark]

Solve for X, Y, Z in: $XYZ + XYZ = 756$ with digits. Find X, Y, Z:

A X=3, Y=7, Z=8**B** X=3, Y=6, Z=8**C** X=4, Y=7, Z=8**D** X=3, Y=8, Z=7**Question 135**

[1 Mark]

In a survey of a town represented by a 3-circle Venn Diagram (Educated, Employed, Backward):

Educated only = 8

Educated & Employed only = 6

All three = 3

Employed only = 7

Backward only = 17

Backward & Employed only = 5

Backward & Educated only = 11

How many educated people are employed?

A 9**B** 6**C** 14**D** 17**Question 136**

[1 Mark]

Using the same Venn Diagram data (Educated only = 8, All three = 3, Backward & Educated only = 11), how many backward people are educated?

A 14**B** 11**C** 9**D** 18**Question 137**

[1 Mark]

Using the same Venn Diagram data, how many backward uneducated people are employed?

A 5**B** 8**C** 11**D** 14**Question 138**

[1 Mark]

Using the same Venn Diagram data, how many backward people are not educated?

A 22**B** 17**C** 20**D** 25

Question 139

[1 Mark]

Eight persons (L, M, N, O, P, Q, R, S) sit in a circle facing inward. N is between L and R. P is second left of N. P and S have 2 persons between them. M neighbors O. One person sits between M and Q. What is O's position relative to S?

☐ A Third to the left☐ B Immediate right☐ C Second to the right☐ D Immediate left**Question 140**

[1 Mark]

Based on the 8-person circular table arrangement (L, M, N, O, P, Q, R, S), four pairs share an adjacent relationship. Which pair does NOT belong to the group?

☐ A RP☐ B QL☐ C NM☐ D MS**Question 141**

[1 Mark]

In the 8-person circular seating arrangement, who sits second to the left of the person immediately to the right of O?

☐ A Q☐ B M☐ C S☐ D L**Question 142**

[1 Mark]

In the 8-person circular seating arrangement, who sits directly between M and Q?

☐ A O☐ B L☐ C N☐ D S**Question 143**

[1 Mark]

In the 8-person circular seating arrangement, who sits third to the right of the person opposite N?

☐ A L☐ B S☐ C R☐ D O

Question 144

[1 Mark]

Seven friends (P, Q, R, S, T, U, V) sit in a circle facing the center. V is 2nd left of S and neighbors T and U. S is not a neighbor of R or T. P neighbors Q and R. Which of the following is correct?

A Q is between P and S**B** S is between U and P**C** T is immediate right of V**D** U is immediate left of V**Question 145**

[1 Mark]

In the 7-friends circular seating arrangement, in which of the following pairs is the second person sitting to the immediate right of the first?

A QU**B** VU**C** TR**D** PT**Question 146**

[1 Mark]

If T and S interchange their places in the 7-friends circle, what will be S's new position?

A Neighbors of V and R**B** Immediate left of R**C** Immediate right of U**D** Neighbors of R and P**Question 147**

[1 Mark]

In the 7-friends circular seating arrangement (with friends seated clockwise as S, U, V, T, R, P, Q facing center), what is the position of R?

A Second to the right of Q**B** Third to the left of U**C** To the immediate right of P**D** Second to the left of V**Question 148**

[1 Mark]

Y is to the East of X, which is to the North of Z. If P is to the South of Z, then in which direction of Y is P located?

A South-West**B** South-East**C** North-West**D** South

Question 149

[1 Mark]

A person walks 6 km South, turns right and walks 4 km, then turns left and walks 6 km. In which direction is the person from the starting point?

A South-West**B** South-East**C** North-West**D** West**Question 150**

[1 Mark]

Rasik walked 20 m North. He turned right and walked 30 m. Then he turned right and walked 35 m. Then he turned left and walked 15 m. Finally, he turned left and walked 15 m. In which direction and how many metres is he from his starting position?

A 45 m East**B** 30 m East**C** 45 m West**D** 30 m West